APIs Diagram

1. Layers

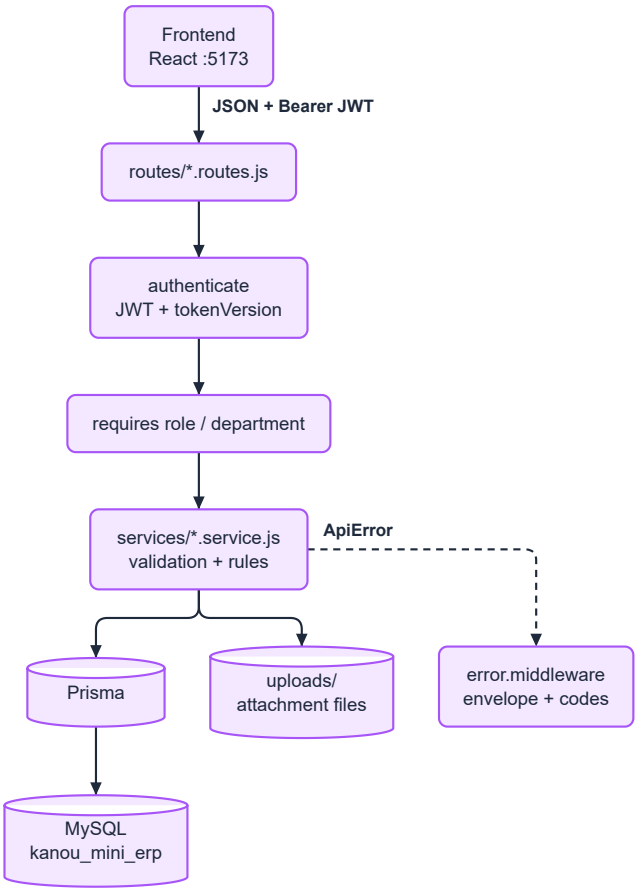


Figure 1. 1. Layers

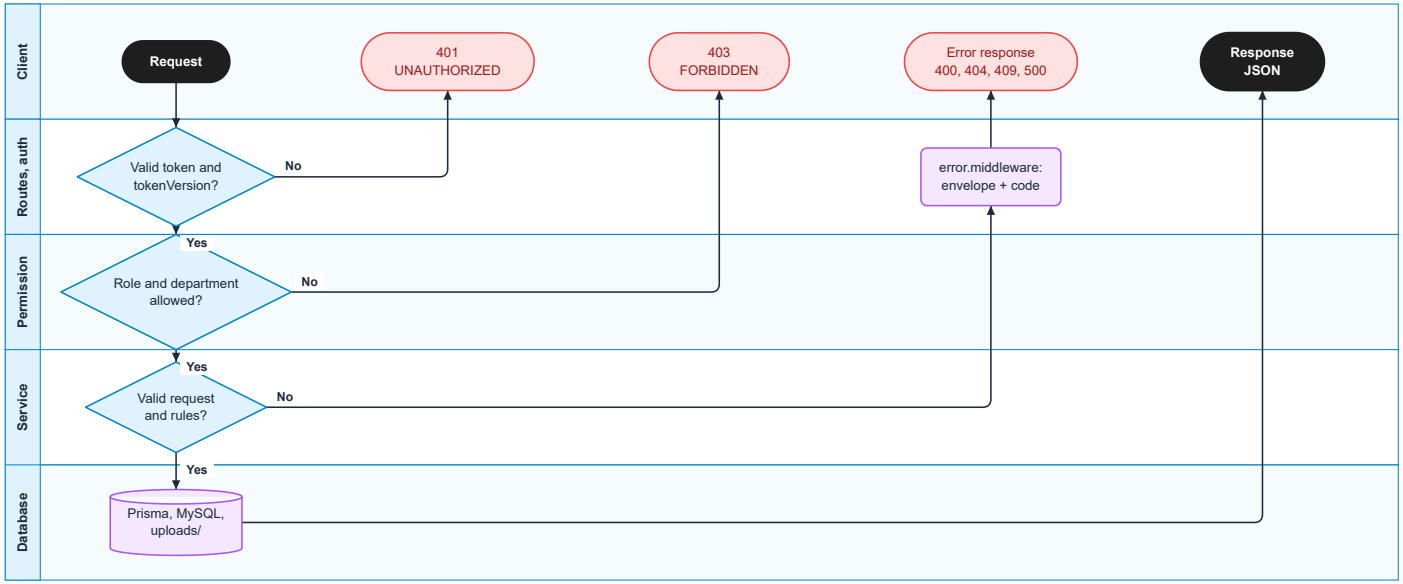


Figure 2. What every API call goes through (swimlanes)

2. Endpoints by module

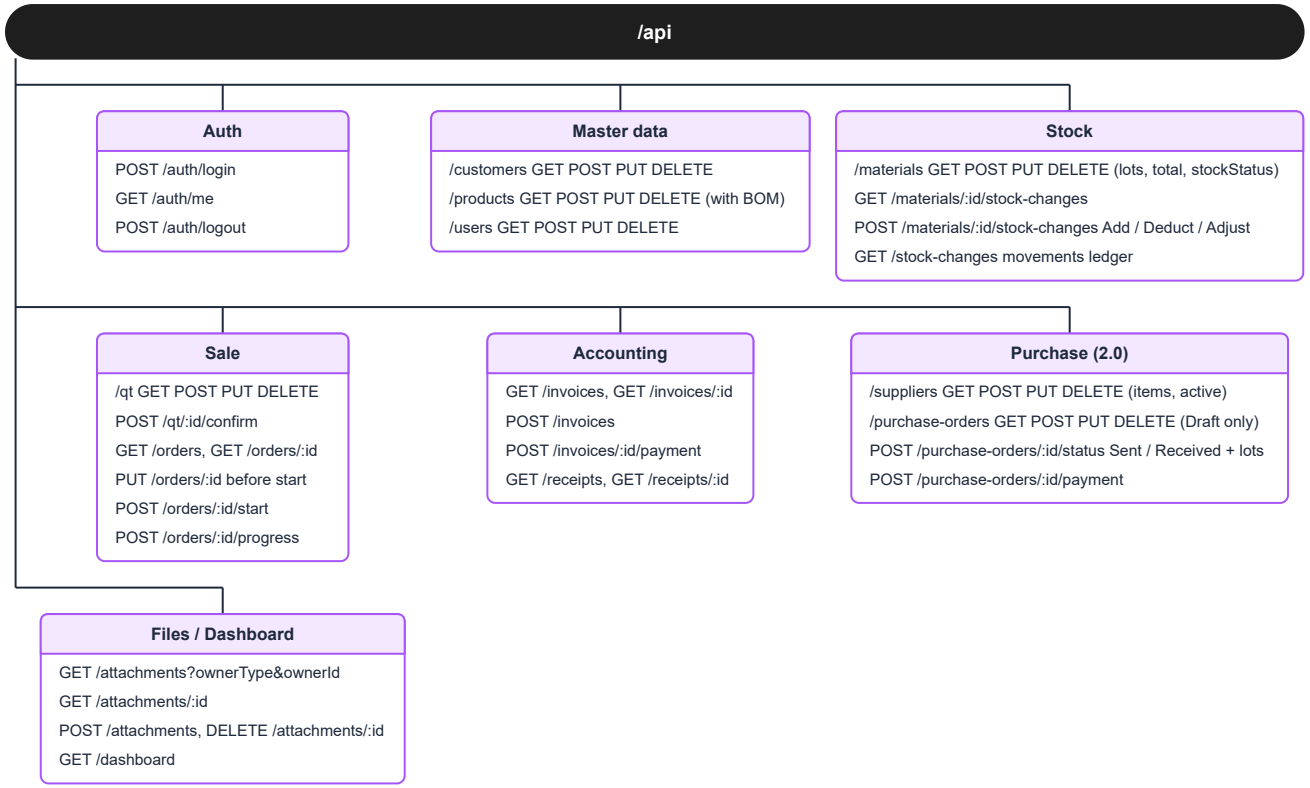


Figure 3. Endpoints by module

Query options added by the merged contract: `GET /qt` takes `step` (`Draft`, `Confirmed`, `InProduction`, `Done`) and every QT has a computed `step`; `GET /orders` takes `invoiced` (`true` / `false`); `GET /suppliers` takes `materialId` and `active`, and a list row has `itemCount` instead of `items`; a purchase order carries `supplierAddress` and its payment is `{ paymentDate, amount, method, referenceNo, note }`.

List calls take `q`, `page`, `pageSize` and answer `meta: { page, pageSize, total }`. `/orders`, `/qt`, `/invoices` also take `status`; `/materials` takes `status` = Normal / Low / Out; `/stock-changes` takes `type`, `from`, `to`.

3. Who may call

Area	Admin	Sale Worker (`Sales`)	Factory Worker (`Factory`)
Dashboard, products read, materials read, stock history, orders read	yes	yes	yes
Customers read	yes	yes	403
Customers, products, materials, users write	yes	403	403
Users read	yes	403	403
QT (all calls, Confirm, delete a Draft)	yes	yes	403
`PUT /orders/:id` (before start)	yes	403	yes
`POST /orders/:id/start`, `/progress`	yes	403	yes
Stock change by hand	yes	403	yes
Reopen / change a Completed process	yes	403	403
Suppliers read; purchase orders, payment, PO files	yes	403	yes
Suppliers write	yes	403	403
Invoices, payments, receipts	yes	403	403
Files of a QT (read, upload, delete)	yes	yes	403
Files of an order: read	yes	yes	yes
Files of an order: upload, delete	yes	403	yes

The role check runs before the record lookup, so a forbidden user gets `403`, not `404`.

4. One sale, end to end

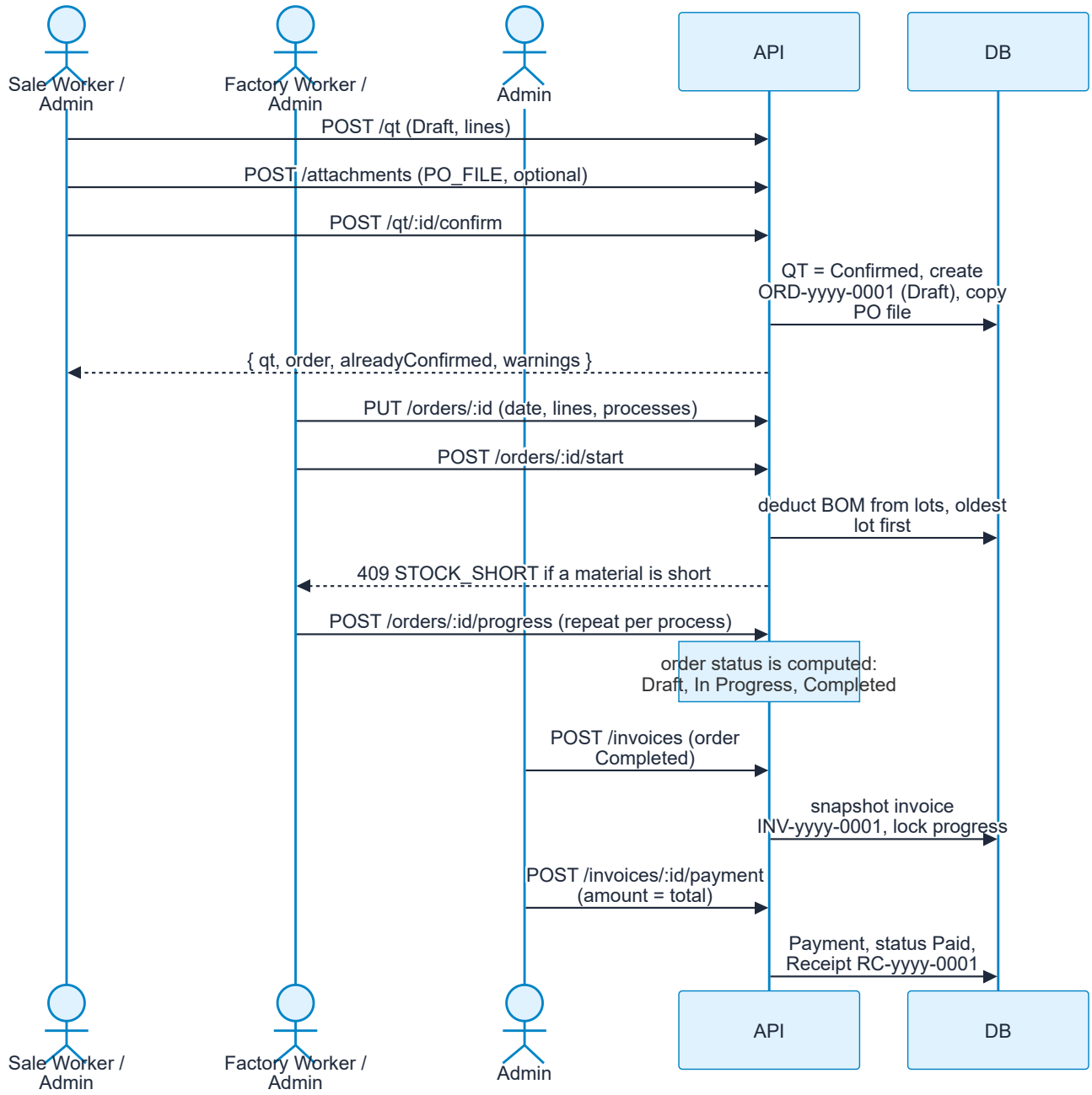


Figure 4. 4. One sale, end to end

Purchase flow

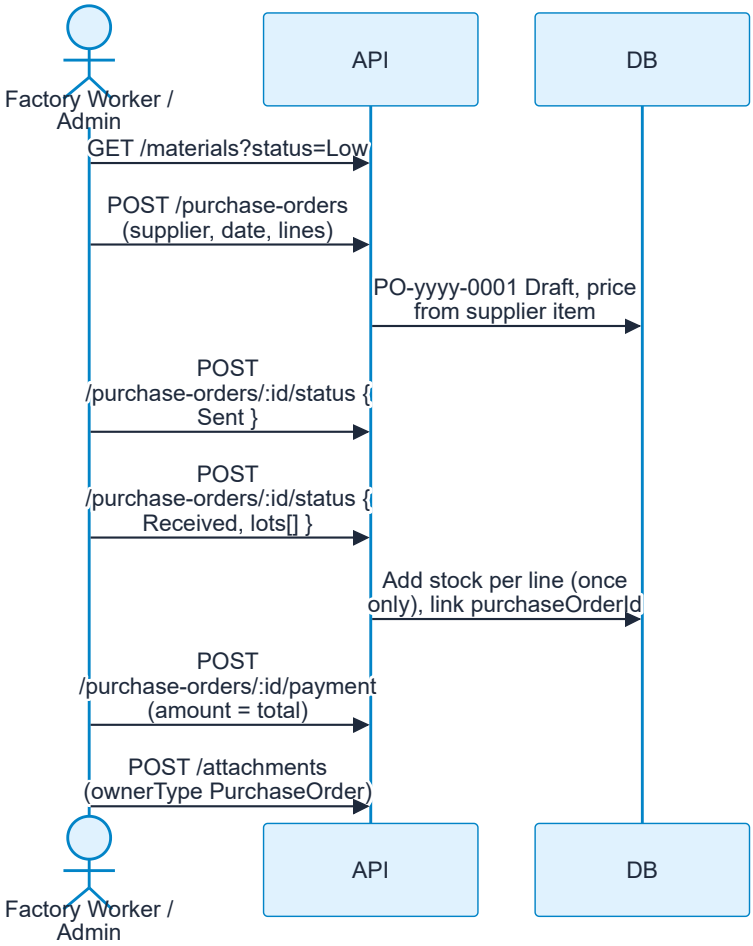


Figure 5. Purchase flow

5. Order and invoice states

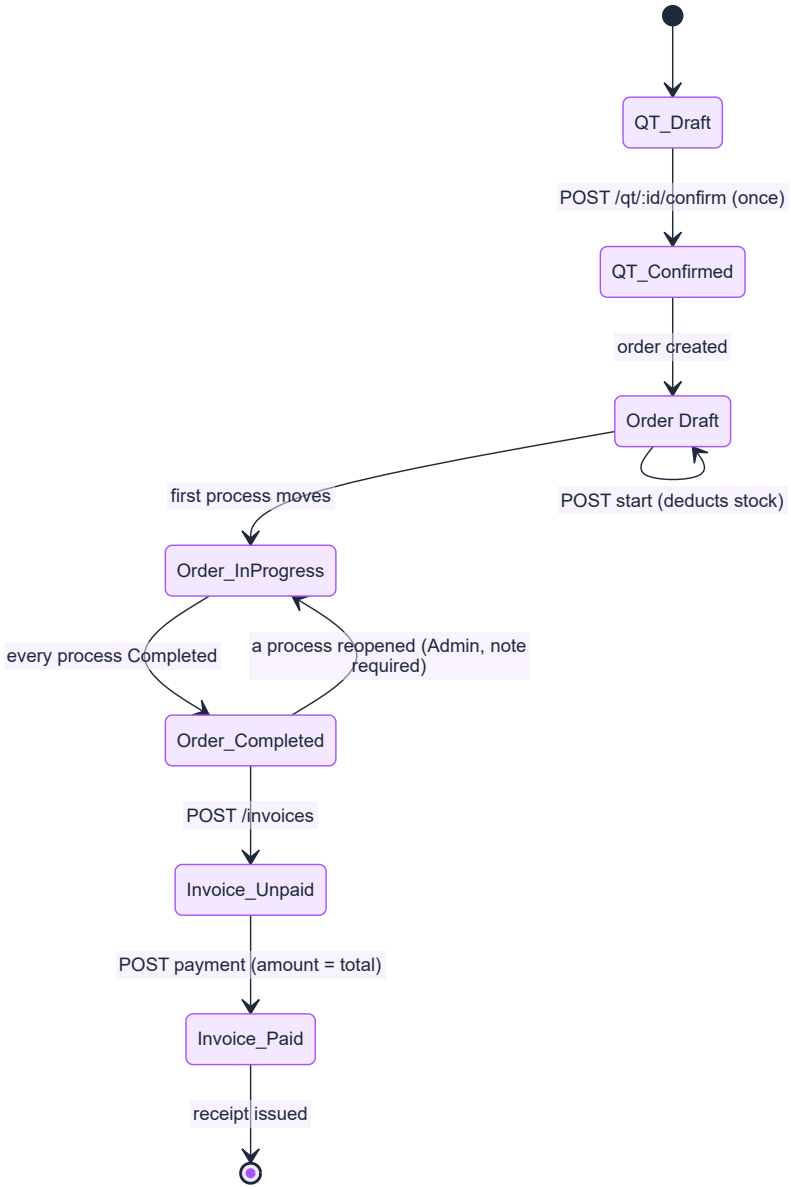


Figure 6. 5. Order and invoice states

6. Data model (tables)

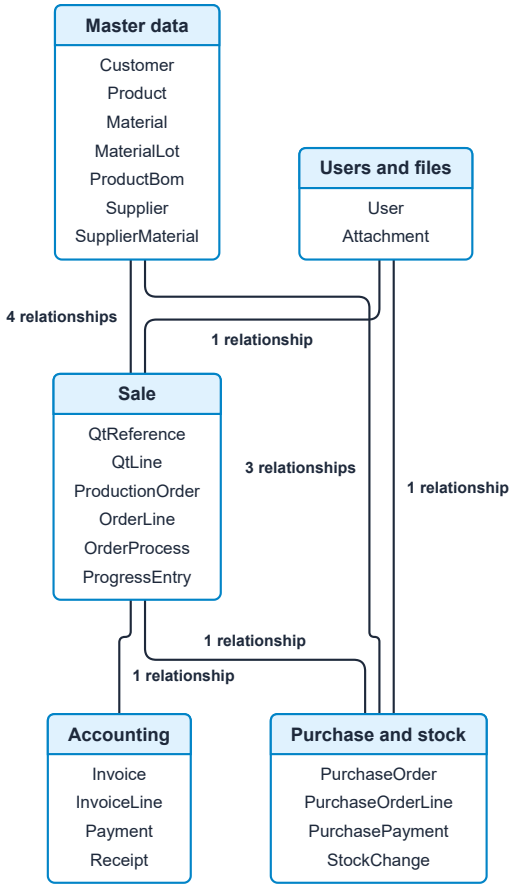


Figure 7. Data model: the 23 tables in 5 groups, and the relationships between the groups

`Attachment` points to its owner (`QtReference` or `ProductionOrder`) by `ownerType` + `ownerId`, with no database foreign key.

7. Error codes

HTTP	`error`	Typical cause
400	`VALIDATION_ERROR`	`errors: [{ path, code }]`, codes `REQUIRED`, `INVALID`, `INVALID_DATE`, `DUPLICATE`, `TOO_SHORT`, `TOO_HIGH`, `MUST_EQUAL_TOTAL`, `FILE_TYPE`, `FILE_TOO_BIG` …
401	`UNAUTHORIZED`, `INVALID_CREDENTIALS`	No / bad / old token; wrong login
403	`FORBIDDEN`, `COMPLETED_LOCKED`	Role or department not allowed
404	`NOT_FOUND`	No such record
409	`DUPLICATE_VALUE`	Unique `code`, `email`, `qtNumber` used
409	`IN_USE`	Delete of a used record; `errors: [{ kind, count, items }]`
409	`STOCK_SHORT`, `STOCK_NEGATIVE`, `LOT_NOT_FOUND`	Stock rules
409	`QT_LOCKED`, `ORDER_LOCKED`, `ALREADY_STARTED`, `ORDER_NOT_READY`, `ORDER_NOT_STARTED`, `ORDER_INVOICED`	Order / QT lifecycle
409	`PO_LOCKED`, `INVALID_STATUS_CHANGE`, `PO_NOT_RECEIVED`	Purchase order lifecycle (`ALREADY_PAID` also covers a PO)
400	`SUPPLIER_INACTIVE`, `NOT_SOLD_BY_SUPPLIER` (codes inside `VALIDATION_ERROR`)	Purchase order lines
409	`ORDER_NOT_COMPLETED`, `INVOICE_EXISTS`, `ALREADY_PAID`, `FILE_LOCKED`	Accounting and files
429	`TOO_MANY_REQUESTS`	Login rate limit
500	`INTERNAL_SERVER_ERROR`	Unexpected failure

8. Tests

`cd backend && pnpm test` runs integration tests (Node test runner) that call the real Express app over HTTP against a separate database, `kanou_mini_erp_test` (made from `DATABASE_URL` with `_test` added; the tests refuse to run on any other name, because they empty every table). The schema is pushed by `test-support/setup-db.js` first.

File	Covers
`test/auth.test.js`	login, `/me`, logout, sessions ended by a new password, rate limit, 401 on every module
`test/master.test.js`	customers, products with BOM, users: validation, search, paging, roles, `IN_USE`, duplicates
`test/stock.test.js`	materials, lots, status, stock changes (Add / Deduct / Adjust), history, movements, delete rules
`test/sale.test.js`	QT, Confirm QT, files (PO copy to the order), production orders, start (stock deduction), progress rules
`test/accounting.test.js`	invoices (snapshot), payment, receipts, dashboard
`test/purchase.test.js`	suppliers, purchase orders, status path, receiving (stock once), payment, files, in-use rules

9. Status

All endpoints above are implemented in `backend/src` and covered by the tests. The development database `kanou_mini_erp` still has to be created with `pnpm run db:setup` (users `admin@gmail.com` / `admin`, `worker@gmail.com` / `worker`, `sale@gmail.com` / `sale`).